MANAS COGNITIVE TRAINING GAMES

Complete Game Design Document (GDD)

Table of Contents

- 1. Signal Spotter
- 2. Story Recall
- 3. Code Cracker
- 4. Travel Bag
- 5. Detective's Notebook
- 6. Sorting Flow
- 7. Time Lock
- 8. Multi-Tasking Challenge
- 9. Endurance Run
- 10. Focus Anchor
- 11. Beat Recall
- 12. Drift Control

Cognitive Domains Summary

This suite covers twelve distinct areas of cognitive function:

Selective Attention	Context Awareness	Deductive Logic
Memory Sequencing	Observation	Cognitive Flexibility

Internal Timing	Task Switching	Persistence
Sustained Attention	Auditory Working Memory	Micro-Control & Impulse Regulation

GAME 1 — SIGNAL SPOTTER

SELECTIVE ATTENTION

1. OVERVIEW

Signal Spotter trains selective attention by requiring users to identify and respond only to target stimuli within a visually noisy environment.

2. OBJECTIVE

Tap only the designated signal while ignoring distractors.

3. CORE MECHANICS

- Targets and distractors appear on screen.
- Only one color/symbol is valid per round.
- Tapping incorrect items reduces accuracy.
- Speed increases gradually.

4. PLAYER FLOW

Intro → Practice Round → 60-second session → Results → Reflection

5. DIFFICULTY SCALING

- Increase spawn speed
- Increase number of distractors
- Reduce target visibility
- Introduce similar-looking decoys

6. SCORING

Score = (Correct taps × speed multiplier) – incorrect penalties. Accuracy % prominently displayed.

7. FAILURE CONDITIONS

Accuracy drops below threshold OR time ends.

8. UI/UX GUIDELINES

Minimalist background, high contrast targets to ensure focus remains on the stimuli.

9. AUDIO DESIGN

Subtle correct/incorrect cues to provide immediate feedback without breaking concentration.

> **10. Data Tracking:** Reaction time, false positives, attention decay curve.

GAME 2 — STORY RECALL

CONTEXT AWARENESS

1. OVERVIEW

Users watch a 40-second animated story and answer recall + inference questions.

2. OBJECTIVE

Correctly answer questions based on details and context.

3. CORE MECHANICS

- Animation plays once.
- 3–5 questions follow immediately after.
- Questions include factual (explicit details) and interpretive (inference based) queries.

4. PLAYER FLOW

Watch → Question set → Results → Reflection

5. DIFFICULTY SCALING

- More subtle details
- Fewer obvious cues
- Multi-character narratives

6. SCORING

Points per correct answer. Bonus awarded for perfect recall.

7. FAILURE CONDITIONS

Score below 50%.

8. UI/UX GUIDELINES

Full-screen cinematic animation player.

9. AUDIO DESIGN

Narrative sound design, distinct voice acting.

> **10. Data Tracking:** Memory accuracy, inference accuracy.

GAME 3 — CODE CRACKER

DEDUCTIVE LOGIC

1. OVERVIEW

Crack a hidden 4–6 digit number using logic feedback.

2. OBJECTIVE

Solve code within limited attempts.

3. CORE MECHANICS

- After each guess, feedback shows:
  - Correct digit & position
  - Correct digit, wrong position
  - Incorrect digit

4. PLAYER FLOW

Guess → Feedback Loop → Solution

5. DIFFICULTY SCALING

- Increase digit length (from 4 to 6)
- Reduce total allowed attempts

6. SCORING

Points based on few attempts used.

7. FAILURE CONDITIONS

Running out of attempts.

8. UI/UX GUIDELINES

Clean, keypad-style interface. Clear visual indicators for feedback states.

9. AUDIO DESIGN

Mechanical clicks, success chime, error buzz.

> **10. Data Tracking:** Attempts used, reasoning efficiency.

GAME 4 — TRAVEL BAG

MEMORY SEQUENCING

1. OVERVIEW

Observe item sequence → reproduce in order.

2. OBJECTIVE

Memorize and re-select items in the exact order they appeared.

3. CORE MECHANICS

- Items appear one by one.
- Player must tap items from a grid in the presented order.

4. PLAYER FLOW

Presentation Phase → Recall Phase → Results

5. DIFFICULTY SCALING

- Start with 3 items, scaling to 8+ items.
- Reduced display time per item.

6. SCORING

Points for correct sequence length.

7. FAILURE CONDITIONS

Incorrect item selection breaks the sequence.

8. UI/UX GUIDELINES

Bright, distinct icons for items. Simple grid layout for recall.

9. AUDIO DESIGN

Distinct tone per item type.

> **10. Data Tracking:** Order accuracy, memory decay.

GAME 5 — DETECTIVE'S NOTEBOOK

OBSERVATION

1. OVERVIEW

Study calm environment → recall object positions.

2. OBJECTIVE

Identify changes or answer specific questions about the scene.

3. CORE MECHANICS

- 20s observation of a detailed scene.
- Scene is hidden.
- User answers specific questions regarding object placement or existence.

4. PLAYER FLOW

Observation → Masking → Query → Results

5. DIFFICULTY SCALING

- More objects in scene.
- Subtle placement differences.

6. SCORING

Accuracy of answers.

7. FAILURE CONDITIONS

Incorrect answers exceeding threshold.

8. UI/UX GUIDELINES

High-resolution artwork, zoom capability.

9. AUDIO DESIGN

Ambient background noise matching the scene context.

> **10. Data Tracking:** Detail recall depth.

GAME 6 — SORTING FLOW

COGNITIVE FLEXIBILITY

1. OVERVIEW

Sort objects by rule; rule changes mid-session.

2. OBJECTIVE

Correctly categorize items despite changing logic.

3. CORE MECHANICS

- User drags items into categories (e.g., Color vs. Shape).
- Rule switch occurs without explicit warning (or very subtle warning).

4. PLAYER FLOW

Sorting Phase 1 → Rule Switch → Sorting Phase 2

5. DIFFICULTY SCALING

- More categories.
- Faster switches between rules.

6. SCORING

Streak bonus for correct sorts after switch.

7. FAILURE CONDITIONS

Too many errors post-switch.

8. UI/UX GUIDELINES

Drag-and-drop interface. Clear feedback on sort success/fail.

9. AUDIO DESIGN

Whoosh sounds for sorting. Glitch sound for rule change.

> **10. Data Tracking:** Adaptation speed post-switch.

GAME 7 — TIME LOCK

INTERNAL TIMING

1. OVERVIEW

User must tap when internal timer reaches exact second (e.g., 10s).

2. OBJECTIVE

Tap exactly when you believe the target time has elapsed.

3. CORE MECHANICS

- No visible countdown is displayed.
- User estimates time internally.

4. PLAYER FLOW

Start Timer → Wait → Tap to Stop → Feedback

5. DIFFICULTY SCALING

- Variable targets (7s, 13s, etc.).
- Distracting background visuals.

6. SCORING

Proximity to target time (milliseconds variance).

7. FAILURE CONDITIONS

Variance > 1.5 seconds.

8. UI/UX GUIDELINES

Abstract, pulsing center circle.

9. AUDIO DESIGN

Metronome sound that fades out after 3 seconds.

> **10. Data Tracking:** Timing deviation accuracy.

GAME 8 — MULTI-TASKING CHALLENGE

TASK SWITCHING

1. OVERVIEW

User tracks 10-second timer while answering interruption question.

2. OBJECTIVE

Answer correctly while stopping the timer accurately.

3. CORE MECHANICS

- Primary task: Track a timer.
- Secondary task: Mid-session pop-up question appears.

- Must resume timer accurately after answering.

4. PLAYER FLOW

Start Task A → Interrupt Task B → Resume Task A

5. DIFFICULTY SCALING

- More complex questions.
- Shorter timing tolerance.

6. SCORING

Composite score of timer accuracy + question accuracy.

7. FAILURE CONDITIONS

Incorrect answer OR poor timing.

8. UI/UX GUIDELINES

Split attention interface. Pop-ups overlaying main task.

9. AUDIO DESIGN

Alert sound for interruption.

> **10. Data Tracking:** Timer accuracy + cognitive recovery speed.

GAME 9 — ENDURANCE RUN

PERSISTENCE

1. OVERVIEW

Tilt/tap to keep character balanced on narrow path.

2. OBJECTIVE

Survive as long as possible.

3. CORE MECHANICS

- Deliberately difficult controls or environment.
- No clear “win” — only time survived.

4. PLAYER FLOW

Start Run → Struggle/Balance → Fail → Restart

5. DIFFICULTY SCALING

- Narrower ledges.
- Environmental instability (wind, shakes).

6. SCORING

Time survived.

7. FAILURE CONDITIONS

Falling off the path.

8. UI/UX GUIDELINES

High-tension visual style. Red/Orange warning indicators.

9. AUDIO DESIGN

Rising tension music. Wind/stress sound effects.

> **10. Data Tracking:** Attempts, time survived, quit rate.

GAME 10 — FOCUS ANCHOR

SUSTAINED ATTENTION

1. OVERVIEW

Maintain focus on central object while distractions increase.

2. OBJECTIVE

React only to changes in the anchor object.

3. CORE MECHANICS

- Tap when subtle anchor change occurs (e.g., slight pulse or color shift).
- Ignore peripheral chaos (moving shapes, flashing lights).

4. PLAYER FLOW

Focus → Identify Change → React → Sustain

5. DIFFICULTY SCALING

- More subtle anchor shifts.
- Faster and brighter distractions.

6. SCORING

Reaction time to anchor changes.

7. FAILURE CONDITIONS

Missed change or reaction to distractor.

8. UI/UX GUIDELINES

Tunnel vision effect. Central element is clear; periphery is blurred/chaotic.

9. AUDIO DESIGN

Steady rhythm (heartbeat) vs chaotic background noise.

> **10. Data Tracking:** Attention stability curve.

GAME 11 — BEAT RECALL

AUDITORY WORKING MEMORY & SEQUENCING

1. OVERVIEW

Beat Recall trains auditory working memory by requiring users to listen to a short rhythmic pattern and reproduce it in the correct order using shuffled beat buttons.

2. OBJECTIVE

Accurately reconstruct the played rhythm sequence.

3. CORE MECHANICS

- A sequence of beats plays automatically.
- Beats vary in tone (e.g., dun, tum, tam).
- After playback, beat buttons appear in random order.
- User taps buttons in the remembered sequence.
- Immediate feedback on submission.

4. PLAYER FLOW

Listen Phase → Reconstruction Phase → Feedback → Results

5. DIFFICULTY SCALING

- Increase sequence length (3 → 8 beats)
- Increase tempo
- Introduce similar-sounding tones
- Add silent beat variation
- Reduce replay opportunities

6. SCORING

Score based on:

- Accuracy (primary)
- Completion speed (secondary)
- Streak bonus for consecutive perfect rounds

7. FAILURE CONDITIONS

- Incorrect sequence
- Timeout (advanced levels only)

8. UI/UX GUIDELINES

- Minimal, dark background
- Subtle pulse animation during playback
- Clear tactile feedback on tap

- Audio-reactive visual glow

9. AUDIO DESIGN

- High-quality, distinct tones
- Clean, premium sound design
- Optional haptic feedback synced to beats

> **10. Data Tracking:** Sequence accuracy, error position (early vs late mistake), reaction time, max sequence length achieved.

GAME 12 — DRIFT CONTROL

MICRO-CONTROL & IMPULSE REGULATION

1. OVERVIEW

Drift Control trains fine motor control and internal stability by requiring users to keep a moving indicator perfectly aligned within a narrow zone using subtle device tilting (gyro-based).

2. OBJECTIVE

Keep the indicator within a target zone for a sustained period without overcorrecting.

3. CORE MECHANICS

- A small orb or line appears in the center of the screen.
- The user tilts their phone to keep it aligned within a narrow boundary.
- The system introduces subtle environmental drift (wind effect).
- Overcorrection causes instability.
- Smooth, minimal movement is rewarded.

4. PLAYER FLOW

Calibration → Stability Phase → Increasing Drift → Results

5. DIFFICULTY SCALING

- Narrower stability zone
- Stronger drift effects

- Longer hold duration
- Introduce visual distractions

6. SCORING

Score based on:

- Stability percentage
- Smoothness of correction
- Time maintained in zone

Bonus for maintaining steady control without sharp movements.

7. FAILURE CONDITIONS

- Indicator leaves boundary for more than X seconds
- Excessive oscillation detected

8. UI/UX GUIDELINES

- Minimalist interface
- Calm gradient background
- Subtle wind or motion cues
- Smooth animations (no arcade feel)

9. AUDIO DESIGN

- Low ambient tone
- Gentle instability sound if drifting
- Calm completion sound

> **10. Data Tracking:** Micro-movement amplitude, overcorrection frequency, stability duration, drift recovery speed.